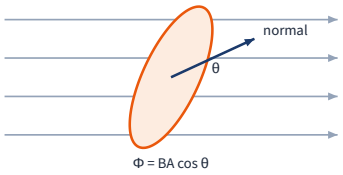
CHAPTER 18

ELECTRO-MAGNETIC INDUCTION

Key idea: a changing flux linkage induces an e.m.f. that opposes the change.

1 MAGNETIC FLUX

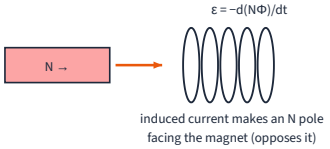


$$\Phi = BA\cos\theta$$

$$\text{linkage} = N\Phi$$

θ between B and the **normal**. 1 Wb = 1 T m².

2 FARADAY'S & LENZ'S LAWS



$$\varepsilon = -N\frac{d\Phi}{dt}$$

- Faraday: $\varepsilon \propto$ rate of change of flux linkage
- Lenz: induced current **opposes** the change (energy conservation)

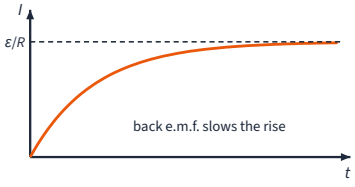
3 MOTIONAL E.M.F.

$$\varepsilon = BLv \qquad F = \frac{B^2L^2v}{R}$$

$$\varepsilon = NBA\omega\sin\omega t$$

Rotating coil (a.c. generator): peak $\varepsilon_0 = NBA\omega$ when the plane is **parallel** to B.

4 SELF-INDUCTION

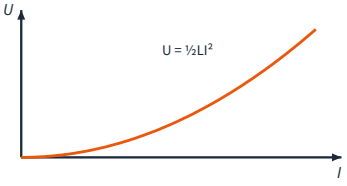


$$\varepsilon = -L\frac{dI}{dt}$$

$$L = \frac{N\Phi}{I} = \frac{\mu_0 N^2 A}{l}$$

Back e.m.f. slows the growth of current; spark when switched off.

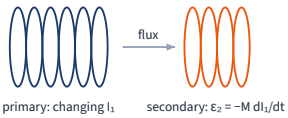
5 ENERGY IN AN INDUCTOR



$$U = \tfrac{1}{2}LI^2 \qquad u = \frac{B^2}{2\mu_0}$$

Stored in the **magnetic field**; returned when the current falls.

6 MUTUAL INDUCTION



$$\varepsilon_2 = -M\frac{dI_1}{dt}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

Transformer works on a.c. only; ideal: $V_pI_p = V_sI_s$.

7 UNIT CHECK

Quantity	Unit	In base form
Flux Φ	Wb	T m² = V s
Flux density B	T	Wb m^{-2}
Inductance L, M	H	Wb A^{-1} = V s A^{-1}
E.m.f.	V	Wb s^{-1}

8 COMMON SECTION A TRAPS

- Taking θ from the plane instead of the normal.
- Forgetting N in flux linkage.
- Peak e.m.f. occurs when the flux is **zero**.
- Saying an e.m.f. needs a closed circuit (only a current does).
- Transformer on d.c.: no output.

EXAM TIPS

- ✓Find $\Delta(N\Phi)$ first, then divide by Δt.
- ✓Explain direction with Lenz's law in words.
- ✓Use $\varepsilon = BLv$ only when B, L, v are perpendicular.
- ✓Energy: work against the opposing force = electrical energy.